

MUSIC STORE DATABASE

Business Requirements Document (BRD)

Customer, Catalogue, Playlist, Order and Payment Data Platform

Version 1.0 | 4 August 2026

Source basis: Music_store_db.sql

Document Control

Field	Value
Document title	Music Store Database - Business Requirements Document
Version	1.0
Status	Design baseline
Prepared for	Music Store Database Design Project
Source schema	Music_store_db.sql
Database platform	PostgreSQL
Schema coverage	14 tables, 93 columns, 16 foreign keys, 9 enum types

Source-derived content reflects the uploaded SQL schema. Items identified as “proposed” or “assumed” are design recommendations that are not explicitly enforced by the current SQL file.

1. Executive Summary

The Music Store Database provides a normalized PostgreSQL foundation for managing registered customers, music metadata, artist and genre relationships, customer playlists, product purchases, order line items, and payment transactions. The design separates catalogue data from customer activity and commerce data while preserving referential integrity through primary keys, foreign keys, unique constraints, enumerated domains, and check constraints.

The database supports the purchase of either individual tracks or complete albums. It also supports many-to-many relationships between artists and albums, artists and tracks, and tracks and genres. Customer playlists retain track order and visibility settings. Orders preserve historical line prices, and payments are recorded separately so that multiple payment attempts can be associated with one order.

2. Business Context and Problem Statement

A digital music store needs a consistent data model that avoids duplicate catalogue records, correctly represents collaborations and classifications, supports customer-curated playlists, and records purchases and payment outcomes without mixing product, order, and payment concepts. The database must also maintain reliable relationships for reporting and future application development.

3. Business Objectives

- Maintain a trusted master catalogue of artists, albums, tracks, genres, and media formats.
- Support artist collaborations and role-based credits at album and track level.
- Allow customers to register, maintain status, and create ordered playlists.
- Support the purchase of individual tracks or complete albums.
- Separate order capture, order line items, and payment transactions.
- Enforce essential data-quality and integrity rules in PostgreSQL.
- Provide a foundation for sales, catalogue, playlist, and payment reporting.

4. Stakeholders

Stakeholder	Interest / Responsibility
Customer	Registers, browses the catalogue, creates playlists, and purchases music.
Catalogue administrator	Maintains artists, albums, tracks, genres, media types, prices, and release metadata.
Commerce administrator	Reviews orders, line items, totals, statuses, and payment outcomes.
Application developer	Builds APIs, user interfaces, and business workflows on the schema.
Database administrator	Deploys, secures, backs up, tunes, and monitors PostgreSQL.
Business analyst / reporting user	Uses the model for customer, catalogue, playlist, sales, and payment analysis.

5. Scope

5.1 In Scope

- Customer identity, location, registration date, and lifecycle status.
- Artist profiles and lifecycle status.
- Albums, album types, prices, release dates, and cover-image references.
- Track metadata, duration, pricing, explicit-content indicator, track/disc sequence, and media format.
- Album-artist, track-artist, and track-genre many-to-many associations.
- Customer playlists, visibility, descriptions, track positions, and added timestamps.
- Order header totals, statuses, currency, and audit timestamps.
- Order items for either tracks or albums.
- Payment amount, status, mode, date, and transaction reference.

5.2 Out of Scope

- Audio file storage, streaming, content delivery, and digital-rights management.
- Shopping cart, wish list, recommendations, reviews, ratings, and social following.
- Discount-code master data, taxation rules, invoicing, and refund allocation details.
- Royalty calculation, licensing, rights ownership, and artist settlement.
- Authentication credentials, passwords, multifactor authentication, and authorization roles.
- Physical inventory, shipping, and fulfilment.

6. High-Level Business Processes

Process	Description
Customer registration	Create a customer with a unique email, registration date, and active status.
Catalogue setup	Create media types, genres, artists, albums, album credits, tracks, track credits, and genre assignments.
Playlist management	Create a customer playlist, define visibility, add tracks,

Process	Description
	and maintain a unique position for each track.
Order capture	Create an order for a customer and calculate subtotal, tax, discount, and total.
Order item capture	Add one or more track or album line items while storing the price charged at purchase time.
Payment processing	Record one or more payment attempts or transactions against an order and maintain status and payment mode.
Reporting	Join customer, catalogue, order, item, and payment data for operational and analytical queries.

7. Functional Requirements

ID	Requirement
FR-01	The system shall create and maintain customer records with unique email addresses.
FR-02	The system shall maintain customer status as active, inactive, or blocked.
FR-03	The system shall maintain artist profiles and artist status.
FR-04	The system shall maintain albums classified as album, EP, single, or compilation.
FR-05	The system shall associate multiple artists with an album and record the artist role.
FR-06	The system shall maintain a unique controlled list of genres.
FR-07	The system shall maintain a unique controlled list of media types.
FR-08	The system shall maintain track metadata, price, duration, media format, explicit-content flag, disc number, and track number.
FR-09	The system shall associate tracks with multiple artists and contribution roles.
FR-10	The system shall associate tracks with multiple genres.
FR-11	The system shall allow a customer to create multiple playlists.
FR-12	The system shall maintain playlist visibility as private, public, or unlisted.
FR-13	The system shall preserve the position and added timestamp of each track in a playlist.
FR-14	The system shall create purchase orders for registered customers.
FR-15	The system shall maintain order status as pending, confirmed, completed, cancelled, or refunded.
FR-16	The system shall store subtotal, tax, discount, total, and currency for each order.
FR-17	The system shall add track or album items to an order.
FR-18	The system shall ensure that an order item references

ID	Requirement
	either a track or an album, not both.
FR-19	The system shall store the unit price and quantity used for each order item.
FR-20	The system shall record one or more payments against an order.
FR-21	The system shall maintain payment status and payment mode using controlled values.
FR-22	The system shall store a unique payment transaction reference when supplied.
FR-23	The system shall provide creation and update timestamps for auditable master and transaction records.

8. Business Rules

ID	Rule
BR-01	Customer email must be unique and not null.
BR-02	Customer status is limited to active, inactive, or blocked.
BR-03	Artist debut year, when supplied, must be at least 1800.
BR-04	Album and track prices cannot be negative.
BR-05	Every track must reference an album and a media type.
BR-06	Track duration, track number, and disc number must be greater than zero.
BR-07	The album, disc number, and track number combination must be unique.
BR-08	Playlist name must be unique within a customer account.
BR-09	Playlist track position must be positive and unique within a playlist.
BR-10	Order subtotal, tax, discount, and total cannot be negative.
BR-11	Order total must equal subtotal plus tax minus discount.
BR-12	An order item must reference exactly one product type: track or album.
BR-13	Order item quantity must be greater than zero.
BR-14	Payment amount must be greater than zero.
BR-15	Payment transaction reference must be unique when present.

9. Data Requirements and Entity Summary

Entity	Purpose	Primary Key
customer	Customer master	customer_id
artist	Artist master	artist_id
album	Album/release catalogue	album_id
album_artist	Album-to-artist association	album_id + artist_id
genre	Music genre master	genre_id

Entity	Purpose	Primary Key
media_type	Audio/media format master	media_type_id
track	Track catalogue	track_id
track_genre	Track-to-genre association	track_id + genre_id
track_artist	Track-to-artist association	track_id + artist_id
playlist	Customer playlist header	playlist_id
playlist_track	Playlist track membership and sequence	playlist_id + track_id
orders	Customer purchase order	order_id
order_item	Purchased track or album line	order_item_id
payment	Order payment transaction	payment_id

10. Entity Relationship Design

The logical ERD below shows the major entities and the associative tables that resolve many-to-many relationships. The separately delivered physical ERD contains all columns and key indicators.

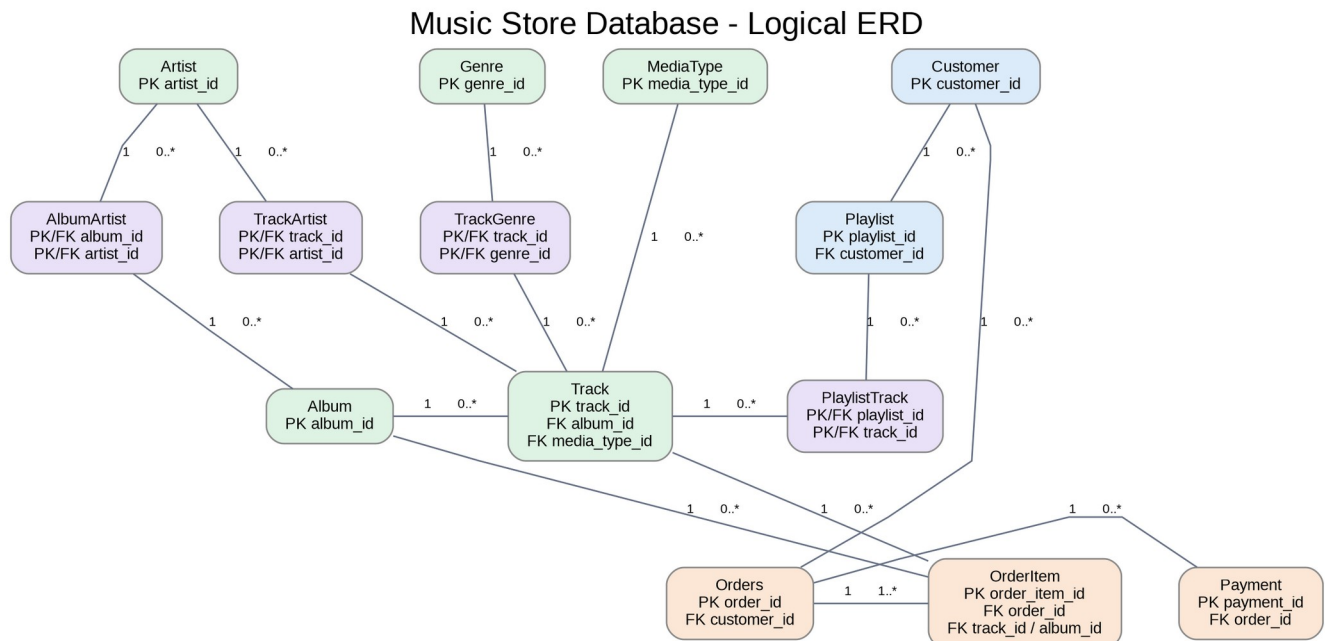


Figure 1. Music Store Database logical entity relationship diagram

11. Non-Functional Requirements

The following are proposed implementation requirements. They are not all explicitly enforced by the current SQL schema.

ID	Category	Requirement
NFR-01	Integrity	PostgreSQL shall enforce primary keys, foreign keys, enums, unique constraints, and check constraints.
NFR-02	Performance	Foreign-key and common search columns shall be indexed based on

ID	Category	Requirement
		query workload.
NFR-03	Security	Application connections shall use least-privilege database roles and encrypted transport.
NFR-04	Privacy	Customer contact data shall be protected and access shall be limited to authorized users.
NFR-05	Auditability	Created and updated timestamps shall be populated consistently; updated_at should be maintained through application logic or a database trigger.
NFR-06	Reliability	Database backups and restore tests shall be scheduled and documented.
NFR-07	Scalability	The design shall support increasing catalogue, playlist, order, and payment volumes without changing core relationships.
NFR-08	Data quality	Email, enum, pricing, quantity, duration, and total validation failures shall be rejected.

12. Proposed Reports and Queries

- Customer registration and status summary by country and city.
- Artist, album, and track catalogue by release type and release date.
- Track catalogue by genre, media type, artist, and explicit-content flag.
- Playlist count and track count by customer and visibility.
- Order value, tax, discount, and status summary by day or customer.
- Top-selling tracks and albums using order-item quantity and unit price.
- Successful, failed, pending, and refunded payments by payment mode.
- Orders with no successful payment or with multiple payment attempts.

13. Acceptance Criteria

ID	Acceptance Criterion
AC-01	All 14 schema tables deploy successfully in PostgreSQL.
AC-02	All primary keys, composite keys, foreign keys, enums, and checks are created.
AC-03	Duplicate customer emails and duplicate controlled names are rejected.
AC-04	A track cannot be inserted without a valid album and media type.
AC-05	Duplicate album/disc/track positions are rejected.
AC-06	A customer can create a playlist and add tracks in unique positive positions.
AC-07	An order item referencing both a track and an album is

ID	Acceptance Criterion
	rejected.
AC-08	An order with an incorrect total calculation is rejected.
AC-09	A positive payment can be associated with a valid order.
AC-10	Core catalogue, playlist, order, and payment joins return consistent results.

14. Assumptions, Limitations, and Design Observations

- The database represents a digital music store and does not store audio binary content.
- Every track belongs to an album record; standalone singles are represented as albums of type single.
- Order currency is stored as a three-character code, but the schema does not validate it against an ISO currency master.
- Tax and discount calculations occur before insert or update; the database only validates the resulting total formula.
- The schema supports multiple payments per order but does not allocate refunds to individual order items.
- The updated_at columns default on insert but require application logic or triggers for later updates.
- The explicit unique index uq_customer_email is redundant because customer.email already has a UNIQUE constraint; it can be removed to avoid duplicate indexing.
- The current composite keys allow one role per artist for each album or track. Multiple simultaneous roles would require artist_role in the primary key or a separate role-assignment table.

15. Requirement-to-Data Traceability

Capability	Tables	Requirements
Customer management	customer	FR-01, FR-02
Artist and album catalogue	artist, album, album_artist	FR-03 to FR-05
Track classification and credits	track, media_type, genre, track_genre, track_artist	FR-06 to FR-10
Playlist management	playlist, playlist_track	FR-11 to FR-13
Order processing	orders, order_item	FR-14 to FR-19
Payment processing	payment	FR-20 to FR-22
Audit timestamps	customer, artist, album, track, playlist, orders, payment	FR-23

Appendix A. Controlled Values

Enum	Allowed Values
customer_status	active, inactive, blocked
artist_status	active, inactive, retired

Enum	Allowed Values
album_type	album, ep, single, compilation
album_artist_role	primary, featured, compilation
track_artist_role	primary, featured, composer, performer, producer
playlist_visibility	private, public, unlisted
order_status	pending, confirmed, completed, cancelled, refunded
payment_status	pending, successful, failed, refunded
payment_mode	card, upi, net_banking, wallet, paypal